$$F^n_{k-1}(\theta,\phi) = \omega \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_{13} e^{-e_{12}\theta} e^{-\phi e_{13} e^{-e_{12}\theta}} e^{-e_{12} n\theta} P^{(n,n)}_{k-n}(\cos\phi)$$

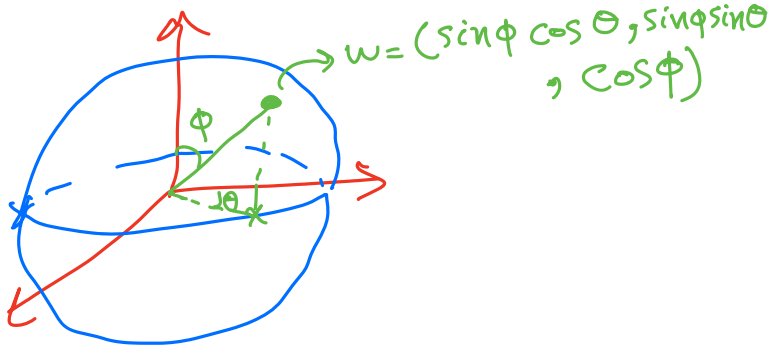
$$- k e_{13} e^{-(n+1)e_{12}\theta} P^{(n,n)}_{k-n-1}(\cos\phi) \Big]. \qquad (3.11)$$

We have this formula for spherical monogenics. To plot we need to obtain the components of F^n_{k-1}, i.e, we have

$$F^n_{k-1}(\theta,\phi) = \left[F^n_{k-1}\right]e_1 + \left[F^n_{k-1}\right]e_2 + \left[F^n_{k-1}\right]e_3 + \left[F^n_{k-1}\right]e_{123}.$$

So $w \in S^2$ so we can write

$$w = \sin\phi\cos\theta\, e_1 + \sin\phi\sin\theta\, e_2 + \cos\phi\, e_3$$



in (3.11) we have $e^{-e_{12}\theta}$ so

$$e^{-e_{12}\theta} = \cos\theta - e_{12}\sin\theta$$

Now

$$e_{13}e^{-e_{12}\theta} = e_{13}(\cos\theta - e_{12}\sin\theta)$$

$$= e_{13}\cos\theta - e_{13}e_{12}\sin\theta$$

$e_1e_3, e_1e_2 = -e_{23}$

$$= e_{13}\cos\theta + e_{23}\sin\theta$$

$$e^{\phi e_{13} e^{-e_{12}\theta}} = \cos\phi + e_{13} e^{-e_{12}\theta} \sin\phi$$

$$e^{\phi e_{13} e^{-e_{12}\theta}} e_3 = e_3 \cos\phi + e_3 (e_{13}\cos\theta + e_{23}\sin\theta)\sin\phi$$

$$= e_3 \cos\phi + e_3 e_{13} \cos\theta \sin\phi + e_3 e_{23} \sin\theta \sin\phi$$

$$= e_3 \cos\phi + e_1 \cos\theta \sin\phi + e_2 \sin\theta \sin\phi$$

$$= w$$

$$w e_{13} e^{-e_{12}\theta} = (e_3 \cos\phi + e_1 \cos\theta \sin\phi + e_2 \sin\theta \sin\phi)(e_{13}\cos\theta + e_{23}\sin\theta)$$

$$= e_3 \cos\phi e_{13}\cos\theta + e_1 \cos\theta \sin\phi e_{13}\cos\theta + e_2 \sin\theta \sin\phi e_{13}\cos\theta + e_3 \cos\phi e_{23}\sin\theta + e_1 \cos\theta \sin\phi e_{23}\sin\theta + e_2 \sin\theta \sin\phi e_{23}\sin\theta$$

$$= e_3 e_{13} \cos\phi \cos\theta + e_1 e_{13} \cos\theta \sin\phi \cos\theta + e_2 e_{13} \sin\theta \sin\phi \cos\theta + e_3 e_{23} \cos\phi \sin\theta + e_1 e_{23} \cos\theta \sin\phi \sin\theta + e_2 e_{23} \sin\theta \sin\phi \sin\theta$$

$$= e_3 e_{13} \cos\phi \cos\theta + e_1 \cos\theta \sin\phi \cos\theta + e_2 \sin\theta \sin\phi \cos\theta + e_3 e_{23} \cos\phi \sin\theta + e_1 \sin\phi \sin^2\theta + e_2

$$= e_1 \cos\theta\cos\phi - \cos^2\theta \sin\phi\, e_3 - e_{123}\sin\theta\cos\theta\sin\phi + e_2\cos\phi\sin\theta + e_{123}\cos\theta\sin\theta\sin\phi - e_3\sin^2\theta\sin\phi$$

$$= -e_3\sin\phi + e_2\cos\phi\sin\theta + e_1\cos\theta\cos\phi$$

$$e^{-\phi e_{13} e^{-e_{12}\theta}} = \cos\phi - e_{13} e^{-e_{12}\theta}\sin\phi$$

$$We_{13}e^{-e_{12}\theta}\; e^{-\phi e_{13} e^{-e_{12}\theta}} = [e_1\cos\theta\cos\phi + e_2\cos\phi\sin\theta - e_3\sin\phi][\cos\phi - e_{13}e^{-e_{12}\theta}\sin\phi]$$

$$= \cos\phi[e_1\cos\theta\cos\phi + e_2\cos\phi\sin\theta - e_3\sin\phi] +$$

$$+\Big[-e_1\cos\theta\cos\phi\, e_{13}\, e^{-e_{12}\theta}\sin\phi$$

$$-e_2\cos\phi\sin\theta\, e_{13}\, e^{-e_{12}\theta}\sin\phi$$

$$+e_3\sin\phi\, e_{13}\, e^{-e_{12}\theta}\sin\phi\Big]$$

$$=\cos\varphi\Big[e_1\cos\theta\cos\phi+e_2\cos\phi\sin\theta$$

$$-e_3\sin\phi\Big]+$$

$$+\Big[\cancel{e_3}\cos\theta\cos\phi\,\cancel{e^{-e_{12}\theta}}\sin\phi$$

$$e_3 e^{-e_{12}\theta}$$

$$-\cancel{e_{13}}\,\cancel{e_2}\, e^{-e_{12}\theta}\cos\phi\sin\theta\sin\phi$$

$$+e_{123}e^{-e_{12}\theta}$$

$$+\cancel{e_{13}}\,\cancel{e_3}\, e^{-e_{12}\theta}\sin^2\phi\Big]$$

$$+e_1 e^{-e_{12}\theta}$$

$$=\cos\varphi\Big[e_1\cos\theta\cos\phi+e_2\cos\phi\sin\theta$$

$$-e_3\sin\phi\Big]+$$

$$\Big[\cos^2\theta \cos\phi \sin\phi \,(e_3 \cos\theta - e_{123} \sin\theta)$$

$$+ \cos\phi \sin^2\theta \sin\phi \,(e_{123} \cos\theta + e_3 \sin\theta)$$

$$+ \sin^2\phi \,(e_1 \cos\theta + e_2 \sin\theta) \Big]$$

$$= \cos\phi \Big[e_1 \cos\theta \cos\phi + e_2 \cos\phi \sin\theta - e_3 \sin\phi \Big] +$$

$$+ \Big[e_3 \cos\phi \sin\phi + e_1 \sin^2\phi \cos\theta + e_2 \sin^2\phi \sin\theta \Big]$$

$$= \Big[e_1 \cos\theta \cos^2\phi + e_2 \cos^2\phi \sin\theta + e_1 \sin^2\phi \cos\theta + e_2 \sin^2\phi \sin\theta \Big]$$

$$= e_1 \cos\theta + e_2 \sin\theta$$

$$\Rightarrow \quad \omega e_{13} e^{-e_{12}\theta} - \phi e_{13} e^{-e_{12}\theta} = e_1 e^{-e_{12}\theta}$$

Jesus!

Therefore,

$$\omega e_{13} e^{-e_{12}\theta} e^{-\phi e_{13} e^{-e_{12}\theta}} e^{-n e_{12}\theta}$$

$$= e_1 e^{-e_{12}\theta} e^{-n e_{12}\theta} e^{-n e_{12}\theta}$$

$$= e_1 e^{-(n+1) e_{12}\theta} = e_1 (\cos(n+1)\theta - e_{12}\sin(n+1)\theta)$$

$$= e_1 \cos(n+1)\theta + e_2 \sin(n+1)\theta$$

$$F_{k-1}^{n}(\theta,\phi) = \omega \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_{13} e^{-e_{12}\theta} e^{-\phi e_{13} e^{-e_{12}\theta}} e^{-e_{12} n\theta} P_{k-n}^{(n,n)}(\cos\phi)$$

$$- k e_{13} e^{-(n+1)e_{12}\theta} P_{k-n-1}^{(n,n)}(\cos\phi) \Big]. \qquad (3.11)$$

$$\Rightarrow F_{k-1}^{n}(\theta,\phi) = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_1 e^{-(n+1)e_{12}\theta} P_{k-n}^{(n,n)}(\cos\phi) - k\omega e_{13} e^{-(n+1)e_{12}\theta} P_{k-n-1}^{(n+1,n)}(\cos\phi) \Big]$$

$$\omega e_{13} e^{-(n+1)e_{12}\theta} = (e_3\cos\phi + e_1\cos\theta\sin\phi + e_2\sin\theta\sin\phi)\, e_{13} e^{-(n+1)e_{12}\theta}$$

$$= (e_1 \cos\phi - e_3 \cos\theta \sin\phi - e_{123} \sin\theta \sin\phi)(\cos(n+1)\theta - e_{12} \sin(n+1)\theta)$$

$e_{123} e_{12} = e_{12} e_{12} e_3 = -e_3$

$$= e_1 \cos\phi \cos(n+1)\theta + e_2 \cos\phi \sin(n+1)\theta - e_3 \cos\theta \cos(n+1)\theta \sin\phi + e_{123} \cos\theta \sin\phi \sin(n+1)\theta - e_{123} \sin\theta \sin\phi \cos(n+1)\theta - e_3 \sin\theta \sin\phi \sin(n+1)\theta$$

$$= e_1 \cos\phi \cos(n+1)\theta + e_2 \cos\phi \sin(n+1)\theta - e_3 \sin\phi (\sin\theta \sin(n+1)\theta + \cos\theta \cos(n+1)\theta) - e_{123} \sin\phi (-\cos\theta \sin(n+1)\theta + \cos(n+1)\theta \sin\theta)$$

$$= e_1 \cos\phi \cos(n+1)\theta + e_2 \cos\phi \sin(n+1)\theta - e_3 \sin\phi (\cos n\theta) + e_{123} \sin n\theta \sin\phi$$

$$= \cos\phi [e_1 \cos(n+1)\theta + e_2 \sin(n+1)\theta] - e_3 \sin\phi \cos n\theta + e_{123} \sin n\theta \sin\phi$$

$$= e_1 \cos\phi \left[e^{-(n+1)\frac{\theta}{2} e_{12}} \right]$$

$$- \sin\phi \left(\cos n\theta - e_{12} \sin n\theta \right) e_3$$

$$= e_1 \cos\phi \, e^{-e_{12}\theta} e^{-n e_{12}\theta}$$

$$= e^{-e_{12} n\theta} \sin\phi \, e_3 \quad \alpha\alpha 2$$

$$F_{k-1}^{n}(\theta,\phi) = \omega \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_{13} e^{-e_{12}\theta} e^{-\phi e_{13} e^{-e_{12}\theta}} e^{-e_{12} n\theta} P_{k-n}^{(n,n)}(\cos\phi)$$

$$- k e_{13} e^{-(n+1)e_{12}\theta} P_{k-n-1}^{(n,n)}(\cos\phi) \Big]. \qquad (3.11)$$

$$\Rightarrow F_{k-1}^{n}(\theta,\phi) = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_1 e^{-(n+1)e_{12}\theta} P_{k-n}^{(n,n)}(\cos\phi) - k\, w e_{13} e^{-(n+1)e_{12}\theta} P_{k-n-1}^{(n,n)}(\cos\phi) \Big]$$

$$= \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_1 e^{-(n+1)e_{12}\theta} P_{k-n}^{(n,n)}(\cos\phi) - k \big[\cos\phi\, e_1 \cos(n+1)\theta + e_2 \cos\phi \sin(n+1)\theta$$

$$-e_3 \cos\phi \sin\phi \cos n\theta$$

$$+ e_{123} \cos\phi \sin n\theta \sin\phi \Big]$$

$$P_{k-n-1}^{(n,n)}(\cos\phi) \Big] \Rightarrow$$

$$\Big[F_{k-1}^{n}(\theta,\phi)\Big]_{e_1} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n)$$

$$\cos(n+1)\theta \; P_{k-n}^{(n,n)}(\cos\phi)$$

$$- k\cos\phi \cos(n+1)\theta \; P_{k-n-1}^{(n,n)}(\cos\phi) \Big]$$

[illegible] by equations [illegible], [illegible], and [illegible], we have

$$F_{k-1}^{n}(\theta,\phi) = \omega \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}} \Big[(k-n) e_{13} e^{-e_{12}\theta} e^{-\phi e_{13} e^{-e_{12}\theta}} e^{-e_{12} n\theta} P_{k-n}^{(n,n)}(\cos\phi)$$

$$- k e_{13} e^{-(n+1)e_{12}\theta} P_{k-n-1}^{(n,n)}(\cos\phi) \Big]. \qquad (3.11)$$

$$\left[F_{k-1}^{n}(\theta,\phi)\right]_{e_2} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[(k-n)\sin(n+1)\theta\, P_{k-n}^{(n,n)}(\cos\phi) - k P_{k-n-1}^{(n,n)}(\cos\phi)(\cos\phi)(\sin(n+1)\theta)\Big]$$

$$F_{k-1}^{n}(\theta,\phi) = \omega\frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[(k-n)e_{13}e^{-e_{12}\theta}e^{-\phi e_{13}e^{-e_{12}\theta}}e^{-e_{12}n\theta}P_{k-n}^{(n,n)}(\cos\phi) - ke_{13}e^{-(n+1)e_{12}\theta}P_{k-n-1}^{(n,n)}(\cos\phi)\Big]. \quad (3.11)$$

$$\left[F_{k-1}^{n}(\theta,\phi)\right]_{e_3} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[k P_{k-n-1}^{(n,n)}(\cos\phi)(\sin\phi)(\sin n\theta)\Big]$$

$$\left[F_{k-1}^{n}(\theta,\phi)\right]_{e_{123}} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[-k P_{k-n-1}^{(n,n)}(\cos\phi)(\sin\phi)(\sin n\theta)\Big]$$

$$[F^n_{k-1}]_{e_1} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[(k-n)P^{(n,n)}_{k-n}(\cos\phi) - k\cos\phi\, P^{(n,n)}_{k-n-1}(\cos\phi)\Big]\cos((n+1)\theta), \tag{3.36}$$

$$[F^n_{k-1}]_{e_2} = \frac{(\sin\phi)^{n-1}}{2^{n+1}\sqrt{\pi}}\Big[(k-n)P^{(n,n)}_{k-n}(\cos\phi) - k\cos\phi\, P^{(n,n)}_{k-n-1}(\cos\phi)\Big]\sin((n+1)\theta), \tag{3.37}$$

$$[F^n_{k-1}]_{e_3} = \frac{k(\sin\phi)^{n}}{2^{n+1}\sqrt{\pi}}P^{(n,n)}_{k-n-1}(\cos\phi)\cos(n\theta), \tag{3.38}$$

$$[F^n_{k-1}]_{e_{123}} = -\frac{k(\sin\phi)^{n}}{2^{n+1}\sqrt{\pi}}P^{(n,n)}_{k-n-1}(\cos\phi)\sin(n\theta). \tag{3.39}$$

$$(2n+\alpha+\beta)(1-z^2)\frac{d}{dz}P^{(\alpha,\beta)}_n(z) = n\big[(\alpha-\beta)-(2n+\alpha+\beta)z\big]P^{(\alpha,\beta)}_n(z) + 2(n+\alpha)(n+\beta)P^{(\alpha,\beta)}_{n-1}(z). \tag{2.9}$$

We shall also use the Jacobi identity

$$qP^{(n,n)}_q(z) - z\left(P^{(n,n)}_q\right)'(z) = \frac{n+q}{2}P^{(n+1,n+1)}_{q-2}(z), \tag{2.11}$$

From (2.11) and $\alpha=\beta=n$ and $n=k-n$, $z=\cos\phi$ we have

$$\Big(2(k-n)+n(n+1)\Big)(1-\cos^2\phi)\Big(P^{(n,n)}_{k-n}\Big)'(\cos\phi)$$

$$= (k-n)\left[(n-n) - (2k-2n+n+1)\cos\phi\right] P_{k-n}^{(n,n)}(\cos\phi)$$

$$+ \; 2(k-n+n)(k-n+n) P_{k-n-1}^{(n,n)}(\cos\phi)$$

$$\Rightarrow (k-n)(-2k)\underbrace{\cos\phi}\, P_{k-n}^{(n,n)}(\cos\phi)$$

$$+ 2k^2 P_{k-n-1}^{(n,n)}(\cos\phi) = 2k \sin^2\phi \left(P_{k-n}^{(n,n)}\right)'(\cos\phi)$$

$$\Rightarrow (k-n)(\cos\phi) P_{k-n}^{(n,n)}(\cos\phi) + \sin^2\phi \left(P_{k-n}^{(n,n)}\right)'(\cos\phi)$$

$$= k P_{k-n-1}^{(n,n)}(\cos\phi)$$

$\times \cos\phi$

$$\Rightarrow (k-n)(\cos\phi)^2 P_{k-n}^{(n,n)}(\cos\phi)$$

$$+ (\sin^2\phi)\cos\phi \left(P_{k-n}^{(n,n)}\right)'(\cos\phi)$$

$$= k(\cos\phi) P_{k-n-1}^{(n,n)}(\cos\phi)$$

So the bracket in (3.36) or (3.37)

$$\left[(k-n)P_{k-n}^{(n,n)}(\cos\phi) - k\cos\phi\, P_{k-n-1}^{(n,n)}(\cos\phi)\right]$$

$$= (k-n)\, P_{k-n}^{(n,n)}(\cos\phi) - (k-n)(\cos^2\phi)\, P_{k-n}^{(n,n)}(\cos\phi)$$

$$- (\sin^2\phi)\underbrace{(\cos\phi)\,(P_{k-n}^{(n,n)})'(\cos\phi)}$$

we use (2.11) with $\ell = k-n$

$$= (k-n)\, P_{k-n}^{(n,n)}(\cos\phi) - (k-n)\cos^2\phi\, P_{k-n}^{(n,n)}(\cos\phi) - \sin^2\phi\left[(k-n)\, P_{k-n}^{(n,n)}(\cos\phi) + \frac{k}{2}\, P_{k-n-2}^{(n+1,n+1)}(\cos\phi)\right]$$

$$= (k-n)\sin^2\phi\, P_{k-n}^{(n,n)}(\cos\phi) - (k-n)\sin^2\phi\, P_{k-n}^{(n,n)}(\cos\phi) + \left(-\frac{k}{2}\right)\sin^2\phi\, P_{k-n-2}^{(n+1,n+1)}(\cos\phi)$$

$$= -\frac{k}{2}\sin^2\phi\, P_{k-n-2}^{(n+1,n+1)}(\cos\phi)$$

Consequently the four new emitted coefficients are

$$[F_{k-1}^n]_{e_1} = -\frac{k}{2^{n+2}\sqrt{\pi}}(\sin\phi)^{n+1} P_{k-n-2}^{(n+1,n+1)}(\cos\phi)\cos((n+1)\theta), \quad (3.41)$$

$$[F_{k-1}^n]_{e_2} = -\frac{k}{2^{n+2}\sqrt{\pi}}(\sin\phi)^{n+1} P_{k-n-2}^{(n+1,n+1)}(\cos\phi)\sin((n+1)\theta), \quad (3.42)$$

$$[F_{k-1}^n]_{e_3} = \frac{k}{2^{n+1}\sqrt{\pi}}(\sin\phi)^{n} P_{k-n-1}^{(n,n)}(\cos\phi)\cos(n\theta), \quad (3.43)$$

$$[F_{k-1}^n]_{e_{123}} = -\frac{k}{2^{n+1}\sqrt{\pi}}(\sin\phi)^{n} P_{k-n-1}^{(n,n)}(\cos\phi)\sin(n\theta). \quad (3.44)$$